

CLAYTON KRISTIANSEN

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TECHNICAL SKILLS

C | C++ | Rust | Assembly | Python | Verilog | Linux | Git | CI/CD | Agile | Test & Verification | Cryptography

EXPERIENCE

Software Engineer II, Texas Instruments

Dallas, TX

Feb 2025 – Present

- Implementing Electrochemical Impedance Spectroscopy (EIS) in C on battery gauges.
 - Executed real-time Discrete Fourier Transforms to process ADC data on limited 16 MHz processor.
- Designed and implemented algorithm to improve gauging accuracy on silicone anode cells.
 - Optimized fixed-point math to ensure rapid, highly accurate execution under strict time constraints.
- Implemented ECC P256 authentication library to be used on all gauge ICs.
 - Achieved < 50ms sign time using precalculated EC random point multiplications in secure memory.
- Built and currently maintaining team's CI/CD process using Jenkins, Docker, and Bitbucket.

Software Engineer I, Texas Instruments

Dallas, TX

June 2023 – Feb 2025

- Developed custom gauge IC firmware utilized in flagship consumer devices for OEMs including Samsung, Dell, Asus, and Lenovo.
- Verified firmware using unit tests written in Python using an ARM simulator.
- Mentored an intern in the creation and verification of an HMAC authentication library.

Digital Design Engineering Intern, Texas Instruments

Federal Way, WA

Jun 2022 – Aug 2022

- Part of a 5-person team using Verilog to validate i2c protocol on one of TI's newest chip designs.
- Created Markdown documentation used by over a dozen engineers throughout the organization.

EDUCATION

Texas A&M University, College Station, Texas

Bachelor of Science in Computer Engineering

GPA: 4.0

Aug 2019 – May 2023

PROJECTS

BQ41Z50 Open Device Partnership Driver (Rust)

Jan 2026 – Present

- Collaborated with Microsoft to create Rust based driver for host devices to communicate with gauge IC.
- Architected overall design and advised Microsoft on changes needed on their side.
- Implemented thread-safe sync. logic to serialize hardware commands and prevent race conditions.
- github.com/OpenDevicePartnership/bq41z50

GraphBLAS (C)

Aug 2022 – May 2023

- Implemented Markov Clustering algorithm using the graphBLAS linear algebra library.
- Contributed to the LAGraph project with Dr. Timothy Davis.
- github.com/GraphBLAS/LAGraph

Buddy Allocator (C++)

Oct 2021 – Dec 2021

- Created a custom memory management system on Linux using buddy allocation.
- Dealt with memory fragmentation by using a slab allocator to break pages down into smaller chunks.
- github.com/claytonkristiansen/BuddyAllocator